A. Stock Arbitraging

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, implementation | https://codeforces.com/problemset/problem/1150/A

Welcome to Codeforces Stock Exchange! We're pretty limited now as we currently allow trading on one stock, Codeforces Ltd. We hope you'll still be able to make profit from the market!

In the morning, there are n opportunities to buy shares. The i-th of them allows to buy as many shares as you want, each at the price of s_i bourles.

In the evening, there are m opportunities to sell shares. The i-th of them allows to sell as many shares as you want, each at the price of b_i bourles. You can't sell more shares than you have.

It's morning now and you possess r bourles and no shares.

What is the maximum number of bourles you can hold after the evening?

Input

The first line of the input contains three integers n, m, r ($1 \le n \le 30$, $1 \le m \le 30$, $1 \le r \le 1000$) — the number of ways to buy the shares on the market, the number of ways to sell the shares on the market, and the number of bourles you hold now.

The next line contains n integers $s_1, s_2, \ldots, s_n$ ($1 \le s_i \le 1000$); s_i indicates the opportunity to buy shares at the price of s_i bourles.

The following line contains m integers $b_1, b_2, \ldots, b_m$ ($1 \le b_i \le 1000$); b_i indicates the opportunity to sell shares at the price of b_i bourles.

Output

Output a single integer — the maximum number of bourles you can hold after the evening.

Examples

Input
3 4 11 4 2 5 4 4 5 4

Output
26

Input
2 2 50 5 7 4 2

Output
50

Note

In the first example test, you have 11 bourles in the morning. It's optimal to buy 5 shares of a stock at the price of 2 bourles in the morning, and then to sell all of them at the price of 5 bourles in the evening. It's easy to verify that you'll have 26 bourles after the evening.

In the second example test, it's optimal not to take any action.